Long Nguyen

LinkedIn | GitHub

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Summary

I am a Software Engineer at Nokia working on AI-RAN, focusing on the L1 aspect of 5G mobile communications. My daily work revolves around real-time digital signal processing, CUDA-accelerated high-performance computing, and modern C++ engineering for mission-critical telecommunications systems.

Experience

Software Engineer — AI-RAN (5G L1) Jan 2026 – Present
Nokia *Espoo, Finland*

- Developing high-performance L1 physical layer components for Nokia's next-generation AI-RAN platform.
- Implementing high-parallel computing kernels in CUDA and C++ for real-time 5G signal processing.
- Optimising latency and throughput constraints in mobile communication systems.
- Skills: CUDA, C++20, 5G NR L1, Parallel Computing, AI-RAN

Trainee — System Modelling & Analytics Jul 2025 – Dec 2025
Nokia *Espoo, Finland*

- Developed visualization tools and automated dashboards for 5G system modelling and performance analysis.
- Integrated continuous integration (CI) workflows and contributed in an Agile Scrum engineering team.
- Skills: CI/CD, Visualization, Scrum, Python

Master's Thesis Researcher Jan 2025 – Jun 2025
Nokia *Espoo, Finland*

- Topic: *5G Physical Layer Channel Conditions Classification Using Machine Learning Techniques*
- Engineered deep learning models for classifying 5G NR TDL/CDL channel conditions and LOS/NLOS detection.
- Conducted rigorous robustness evaluation under severe domain shift in wireless propagation environments.
- Skills: TensorFlow, Wireless AI, 5G NR, Deep Learning

Software Engineering Project — Solibri Autorun Jan 2024 – May 2024
Solibri Company Project *Tampere University*

- Engineered a custom Java Spring Boot desktop interface eliminating XML configuration errors for BIM quality check pipelines.
- Implemented dynamic directory scanning, automated input validation, and intuitive Figma-prototyped UI.
- Recognised with a €3 000 Project Scholarship for outstanding engineering and delivery.
- Skills: Java, Spring Boot, SonarQube, Agile Scrum

Teaching Assistant — Programming 3 (OOP & Java) Sep 2023 – May 2024
Tampere University *Tampere, Finland*

- Instructed lab sessions (Coditoriums) and assisted undergraduate students in advanced Java, API design, and design patterns.
- Evaluated software projects, provided code reviews, and fostered clean architectural habits.
- Skills: Java, APIs, OOP Design, Mentorship

Research Trainee — Light Shaping & Adaptive Optics Jan 2023 – May 2023
Tampere University *Tampere, Finland*

- Applied Zernike polynomials and 4F optical system to a Spatial Light Modulator (SLM) for phase modulation.
- Implemented genetic algorithms in MATLAB to optimize and compensate for optical beam imperfections.
- Skills: MATLAB, Genetic Algorithms, Photonics, Wavefront Optics

Education

Aalto University Espoo, Finland
Master of Science (Technology) *Aug 2024 – Dec 2026*

- GPA: GPA 4.0 / 5.0 (75 ECTS)
- Specialization: Machine Learning, Data Science & AI
- Thesis: *Quantitative RL Trading System — Reinforcement Learning for Algorithmic Trading*

Skills

Programming Languages	:	Python, C++, C, Java, MATLAB, Scala, SQL
Frameworks	:	TensorFlow, PyTorch, Reinforcement Learning (PPO/SAC), Deep Learning, scikit-learn, Computer Vision
Systems & HPC	:	CUDA, High-Parallel Computing, 5G NR L1, Apache Spark, Linux, Docker
Dev Tools	:	FastAPI, Spring Boot, Git / GitHub Actions, SonarQube, MongoDB, SQLite
Pace	:	244+ ECTS / 2 years (Normal Pace: 120 ECTS / 2 years)

Projects

Quantitative RL Trading System 2025 – Present

- Master's thesis research framework developed at Aalto University for building, training, replaying, and inspecting reinforcement-learning trading policies (PPO / SAC). Features timestamped market-data pipelines, causal multi-timeframe feature calculations, broker-neutral interfaces, simulated execution, and rigorous evaluation with full experiment management.
- Skills: Python, Reinforcement Learning, PPO / SAC, Market Microstructure

5G NR Physical Layer Channel Classification Jan 2025 – Jun 2025

- Open research framework for 5G NR physical layer channel analysis and classification (TDL / CDL channel condition models). Investigated physically grounded LOS/NLOS detection under severe domain shift in wireless propagation environments to support adaptive L1 baseband processing.
- Skills: Python, TensorFlow, 5G NR L1, Wireless AI

Solibri Autorun Custom View Tool Jan 2024 – May 2024

- Engineered a desktop application with Java Spring Boot and JavaFX designed to eliminate syntax and semantic errors in complex XML automated building information modeling (BIM) model checking. Featured dynamic directory scanning, automated input constraint validation, Figma UI design, and SonarQube code quality assurance.
- Skills: Java, Spring Boot, JavaFX, XML
- Scholarship of the project: €3 000 Scholarship Award

Transforming Frontal to Lateral: Chest X-ray View Conversion Dec 2023 – Jan 2024

- Synthesised lateral chest radiographs from frontal views using a custom CycleGAN framework with UNet–ResNet architecture and Self-Organised Operational Layers (SOOL). Overcame cross-projection ambiguities in medical imaging to enhance anatomical and diagnostic fidelity.
- Skills: TensorFlow, CycleGAN, Medical Imaging, ResNet

Certifications & Achievements

- **100% Tuition Fee Scholarship**:
- **€3 000 Project Scholarship**:
- **MATLAB Fundamentals Certification**:
- **Deep Learning Specialization**:
- **Machine Learning Specialization**:
- **Accelerated Academic Record**:

Additional

Languages : English (Professional), Vietnamese (Bilingual), Finnish (Elementary)